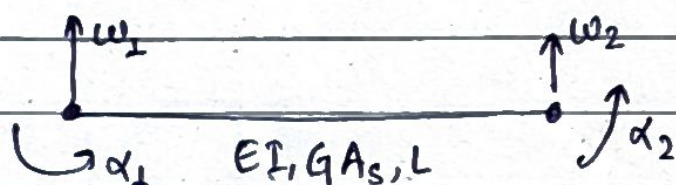


Homework: 3

1

- ① Redo the derivation followed in the class and find the 4×4 element stiffness matrix for a 2D linear-elastic displacement-based Timoshenko beam element considering linear shape functions for vertical displacement and section rotation.



→ Sol?

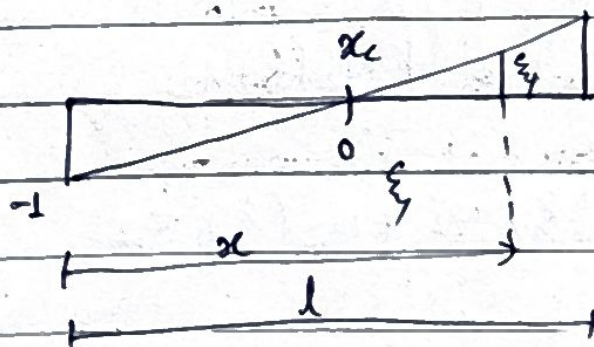
Each node has 2DOF; transverse displacement w , rotation α for the given element,

$$\text{Displacement vector, } \underline{\tilde{u}}(x) = \begin{bmatrix} \alpha(x) \\ w(x) \end{bmatrix}$$

The internal displacement vector has two fields $w(x)$ and $\alpha(x)$, and are completely separate (Timoshenko Beam).

$$\text{The nodal displacement vector, } \underline{\tilde{u}} = \begin{bmatrix} w_1 \\ \alpha_1 \\ w_2 \\ \alpha_2 \end{bmatrix}$$

Mapping physical coordinate x to natural coordinate $\xi \in [-1, 1]$,



$$\frac{\xi - 1}{2} = \frac{x - l}{l}$$

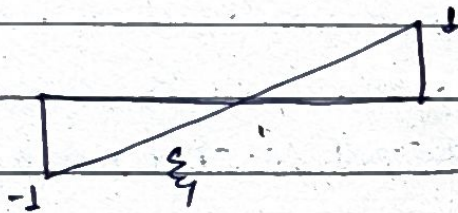
$$\Rightarrow \xi = \frac{2}{l} \left(x - \frac{l}{2} \right)$$

$$\Rightarrow \xi = \frac{2x}{l} - 1$$

$$\Rightarrow \frac{d\xi}{dx} = \frac{2}{l}$$

$$\text{Jacobian } (J) = \frac{dx}{d\xi} = \frac{l}{2}$$

① Shape Functions,



Using Lagrange Polynomial,

$$l(x) = \prod_{\substack{m=1 \\ m \neq k}}^n \frac{x - x_m}{x_k - x_m}$$

for two noded element,

$$N_1(\xi) = \frac{\xi - 1}{-1 - 1} = \frac{1 - \xi}{2}$$

$$N_2(\xi) = \frac{\xi + 1}{1 + 1} = \frac{1 + \xi}{2}$$

} linear shape functions for $w(\xi)$ and $\alpha(\xi)$

Thus,

internal displacement fields are,

$$\alpha(\xi) = N_1(\xi) \alpha_1 + N_2(\xi) \alpha_2$$

$$w(\xi) = N_1(\xi) w_1 + N_2(\xi) w_2$$

In Matrix form,

$$\underline{\underline{u}}(x) = \begin{bmatrix} \alpha(x) \\ \omega(x) \end{bmatrix} \Rightarrow \underline{\underline{u}}(\xi) = \begin{bmatrix} \alpha(\xi) \\ \omega(\xi) \end{bmatrix}$$

$$\underline{\underline{u}}(\xi) = \begin{bmatrix} \alpha(\xi) \\ \omega(\xi) \end{bmatrix} = \begin{bmatrix} 0 & N_1 & 0 & N_2 \\ N_1 & 0 & N_2 & 0 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \alpha_1 \\ \omega_2 \\ \alpha_2 \end{bmatrix}$$

$$\Rightarrow \underline{\underline{u}}(\xi) = \underline{\underline{N}} \underline{\underline{u}}$$

(II) Compatibility

Corresponding strains for displacement fields,

$$\alpha(x) \longrightarrow \kappa(x)$$

$$\omega(x) \longrightarrow \gamma(x)$$

Bending curvature,

$$\begin{aligned} \kappa(x) &= \frac{d\alpha}{dx} = \frac{d\alpha}{d\xi} \frac{d\xi}{dx} = \left(\frac{2}{l}\right) \frac{d\alpha}{d\xi} = \left(\frac{2}{l}\right) \left(N_1' \alpha_1 + N_2' \alpha_2 \right) \\ &= \left(\frac{2}{l}\right) \left(-\frac{1}{2} \alpha_1 + \frac{1}{2} \alpha_2 \right) = \frac{1}{l} (-\alpha_1 + \alpha_2) \end{aligned}$$

$$\Rightarrow \kappa(x) = \kappa(\xi) = \frac{-\alpha_1}{l} + \frac{\alpha_2}{l}$$

Shear strain,

$$\gamma(x) = \alpha(x) - \frac{d\omega(x)}{dx}$$

$$\Rightarrow \gamma(\xi) = \alpha(\xi) - \frac{d\omega(\xi)}{d\xi} \frac{d\xi}{dx}$$

$$\Rightarrow \gamma(\xi) = \alpha(\xi) - \left(\frac{2}{l}\right) \frac{dw}{d\xi}$$

$$= \frac{1-\xi}{2} \alpha_1 + \frac{1+\xi}{2} \alpha_2 - \frac{2}{l} \left(-\frac{l}{2} \omega_1 + \frac{l}{2} \omega_2 \right)$$

$$\Rightarrow \gamma(\xi) = \frac{1}{l} \omega_1 + \frac{1-\xi}{2} \alpha_1 - \frac{1}{l} \omega_2 + \frac{1+\xi}{2} \alpha_2$$

Thus strain vector,

$$\underline{\epsilon}(\xi) = \begin{bmatrix} \kappa(\xi) \\ \gamma(\xi) \end{bmatrix} = \begin{bmatrix} 0 & -1/l & 0 & 1/l \\ 1/l & \frac{1-\xi}{2} & -1/l & \frac{1+\xi}{2} \end{bmatrix} \underline{u}$$

$$\Rightarrow \underline{\epsilon}(\xi) = \underline{B} \underline{u}$$

(iii) Constitutive

Corresponding stress vector,

$$\underline{\sigma}(x) = \begin{bmatrix} M(x) \\ V(x) \end{bmatrix}$$

we have,

$$M(x) = EI \kappa(x) \Rightarrow M(\xi) = EI \kappa(\xi)$$

$$V(x) = GA_s \gamma(x) \Rightarrow V(\xi) = GA_s \gamma(\xi)$$

In matrix form,

$$\Rightarrow \underline{\sigma}(\xi) = \begin{bmatrix} EI & 0 \\ 0 & GA_s \end{bmatrix} \begin{bmatrix} \kappa(\xi) \\ \gamma(\xi) \end{bmatrix}$$

$$\Rightarrow \underline{\sigma}(\xi) = \underline{k} \underline{B} \underline{u}$$

(iv) Element stiffness matrix,

$$\underline{\underline{K_e}} = \int_{-1}^1 \underline{\underline{B}}^T \underline{\underline{K}} \underline{\underline{B}} d\xi J = \frac{1}{2} \int_{-1}^1 \underline{\underline{B}}^T \underline{\underline{K}} \underline{\underline{B}} d\xi \quad \left(\text{since } J = \frac{L}{2} \right)$$

we have,

$$\underline{\underline{K}} \underline{\underline{B}} = \begin{bmatrix} EI & 0 \\ 0 & GA_s \end{bmatrix}_{2 \times 2} \begin{bmatrix} 0 & -1/L & 0 & 1/L \\ 1/L & \frac{1-\xi}{2} & -1/L & \frac{1+\xi}{2} \end{bmatrix}_{2 \times 4}$$

$$= \begin{bmatrix} 0 & -EI/L & 0 & EI/L \\ \frac{GA_s}{L} & \frac{GA_s(1-\xi)}{2} & -\frac{GA_s}{L} & \frac{GA_s(1+\xi)}{2} \end{bmatrix}$$

$$\underline{\underline{B}}^T \underline{\underline{K}} \underline{\underline{B}} = \begin{bmatrix} 0 & 1/L \\ -1/L & (1-\xi)/2 \\ 0 & -1/L \\ 1/L & (1+\xi)/2 \end{bmatrix}_{4 \times 2} \begin{bmatrix} 0 & -EI/L & 0 & EI/L \\ \frac{GA_s}{L} & \frac{GA_s(1-\xi)}{2} & -\frac{GA_s}{L} & \frac{GA_s(1+\xi)}{2} \end{bmatrix}_{2 \times 4}$$

$$\Rightarrow \underline{\underline{B}}^T \underline{\underline{K}} \underline{\underline{B}} = \begin{bmatrix} \frac{GA_s}{L^2} & -\frac{GA_s(\xi-1)}{2L} & -\frac{GA_s}{L^2} & \frac{GA_s(\xi+1)}{2L} \\ \frac{EI}{L^2} + \frac{GA_s(1-\xi)^2}{4} & \frac{GA_s(\xi-1)}{2L} & -\frac{EI}{L^2} - \frac{GA_s(\xi^2-1)}{4} & \frac{GA_s(1-\xi)}{2L} \\ \frac{GA_s}{L^2} & -\frac{GA_s(1+\xi)}{2L} & -\frac{EI}{L^2} - \frac{GA_s(\xi^2-1)}{4} & \frac{GA_s(\xi+1)}{2L} \\ \frac{EI}{L^2} + \frac{GA_s(1+\xi)^2}{4} & \frac{GA_s(\xi+1)}{2L} & \frac{GA_s(1+\xi)}{2L} & \frac{GA_s}{L^2} \end{bmatrix}$$

Symmetric.

Now,

$$K_{11} = \frac{1}{2} \int_{-1}^1 \frac{GA_s}{l^2} d\xi = \frac{1}{2} \left(\frac{GA_s}{l^2} \right) [\xi]_{-1}^1 = \frac{GA_s}{l}$$

$$K_{12} = \frac{1}{2} \int_{-1}^1 \frac{-GA_s}{2l} (\xi-1) d\xi = \frac{-GA_s}{4} \left(\frac{\xi^2}{2} - \xi \right) \Big|_{-1}^1 = \frac{GA_s}{2}$$

Similarly,

$$K_{13} = \frac{1}{2} \int_{-1}^1 \frac{-GA_s}{l^2} d\xi = -\frac{GA_s}{l}$$

$$K_{14} = \frac{1}{2} \int_{-1}^1 \frac{GA_s}{2l} (\xi+1) d\xi = \frac{GA_s}{4} \times 2 = \frac{GA_s}{2}$$

$$\begin{aligned} K_{22} &= \frac{1}{2} \int_{-1}^1 \left(\frac{EI}{l^2} + \frac{GA_s}{4} (1-\xi)^2 \right) d\xi = \frac{EI}{2l} \times 2 + \frac{GA_s l}{8} \times \frac{8}{3} \\ &= \frac{EI}{l} + \frac{GA_s l}{3} \end{aligned}$$

$$K_{23} = \frac{1}{2} \int_{-1}^1 \frac{GA_s}{2l} (\xi-1) d\xi = \frac{GA_s}{4} \times -2 = -\frac{GA_s}{2}$$

$$\begin{aligned} K_{24} &= \frac{1}{2} \int_{-1}^1 \left(\frac{-EI}{l^2} + \frac{GA_s}{4} (\xi^2-1) \right) d\xi = \frac{-EI}{2l} \times 2 - \frac{GA_s l}{8} \times \frac{4}{3} \\ &= -\frac{EI}{l} + \frac{GA_s l}{6} \end{aligned}$$

$$K_{33} = \frac{1}{2} \int_{-1}^1 \frac{GA_s}{l^2} d\xi = \frac{GA_s}{l}$$

$$K_{34} = \frac{1}{2} \int_{-1}^1 \frac{-GA_s}{2l} (1+\xi) d\xi = -\frac{GA_s}{2}$$

$$K_{44} = \frac{1}{2} \int_{-1}^1 \left(\frac{EI}{l^2} + \frac{GA_s}{4} (1+\xi)^2 \right) d\xi = \frac{EI}{l} + \frac{GA_s l}{3}$$

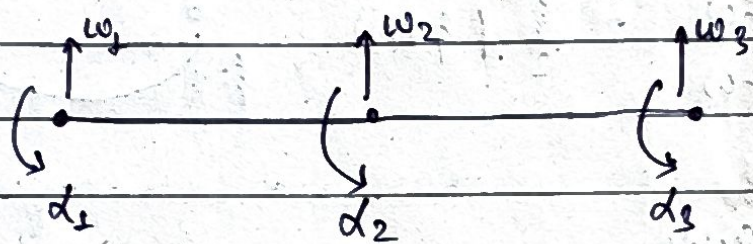
In Matrix form,

$$K_e = \begin{bmatrix} \frac{GA_s}{l} & \frac{GA_s}{2} & -\frac{GA_s}{l} & \frac{GA_s}{2} \\ & \frac{EI}{l} + \frac{GA_s l}{3} & -\frac{GA_s}{2} & \frac{-EI}{l} + \frac{GA_s l}{6} \\ & & \frac{GA_s}{l} & -\frac{GA_s}{2} \\ \text{Sym.} & & & \frac{EI}{l} + \frac{GA_s l}{3} \end{bmatrix}$$

factoring $\phi = \frac{l^2 EI}{GA_s l^2}$ gives,

$$K_e = \frac{GA_s l}{l^2} \begin{bmatrix} l^2/l^2 & 6/l & -l^2/l^2 & 6/l \\ & 4+\phi & -6/l & 2-\phi \\ & & l^2/l^2 & -6/l \\ \text{Sym.} & & & 4+\phi \end{bmatrix}, \phi = \frac{l^2 EI}{GA_s l^2}$$

- ② Redo Problem #1 considering a 3-node finite element to find 6x6 element stiffness matrix. Note that the shape functions will be quadratic for vertical displacement and section rotation. Explain why the stiffness matrix derived in Problem #1 and #2 and the analytically-derived Timoshenko beam stiffness matrix are different.



EI, GA, L

→ Solⁿ. following solution of question ①,

Nodal displacement vector, $\underline{u} =$

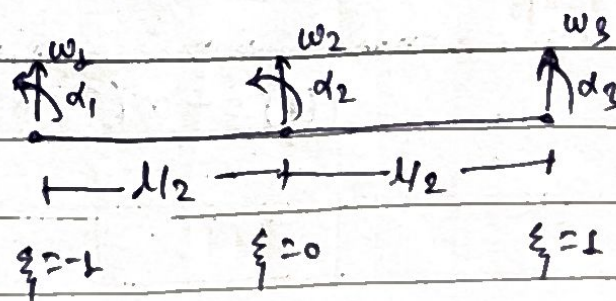
$$\begin{bmatrix} w_1 \\ \alpha_1 \\ w_2 \\ \alpha_2 \\ w_3 \\ \alpha_3 \end{bmatrix}$$

Displacement vector, $u(x) = \begin{bmatrix} \alpha(x) \\ w(x) \end{bmatrix}$

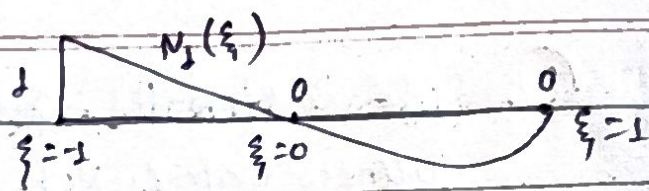
① Shape functions,

using Lagrange polynomial,

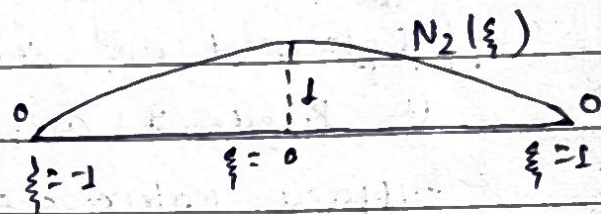
$$L(x) = \prod_{\substack{m=1 \\ m \neq k}}^n \frac{x - x_m}{x_k - x_m}$$



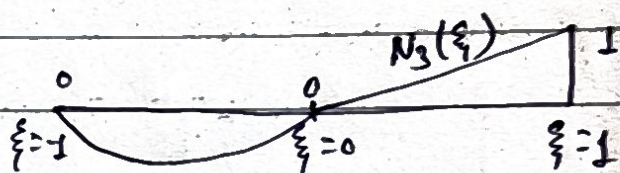
$$N_1 = \frac{(\xi-0)(\xi-1)}{(1-0)(1-1)} = \frac{\xi}{2}(\xi-1)$$



$$N_2 = \frac{(\xi+1)(\xi-1)}{(0+1)(0-1)} = 1-\xi^2$$



$$N_3 = \frac{(\xi+1)(\xi-0)}{(1+1)(1-0)} = \frac{\xi}{2}(\xi+1)$$



Internal displacement fields,

$$w(\xi) = N_1(\xi)w_1 + N_2(\xi)w_2 + N_3(\xi)w_3$$

$$\alpha(\xi) = N_1(\xi)\alpha_1 + N_2(\xi)\alpha_2 + N_3(\xi)\alpha_3$$

$$U(\xi) = \begin{bmatrix} \alpha(\xi) \\ w(\xi) \end{bmatrix} = \begin{bmatrix} 0 & N_1 & 0 & N_2 & 0 & N_3 \\ N_1 & 0 & N_2 & 0 & N_3 & 0 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ w_1 \\ w_2 \\ w_3 \end{bmatrix}$$

(ii) Compatibility,

$$\begin{aligned} K(x) &= \frac{d\alpha(x)}{dx} = \left(\frac{2}{l}\right) \left(N_1'(\xi)\alpha_1 + N_2'(\xi)\alpha_2 + N_3'(\xi)\alpha_3 \right) \\ &= \frac{2}{l} \left(\left(\xi - \frac{1}{2}\right)\alpha_1 - 2\xi\alpha_2 + \left(\xi + \frac{1}{2}\right)\alpha_3 \right) \end{aligned}$$

$$\gamma(\xi) = -\alpha(\xi) + \frac{dw(\xi)}{d\xi} \frac{d\xi}{dx}$$

$$\Rightarrow \gamma(\xi) = -\alpha(\xi) + \left(\frac{2}{l}\right) \frac{d\omega(\xi)}{d\xi}$$

$$= \left(\frac{\xi^2 - \xi}{2}\right) \alpha_1 + (1 - \xi^2) \alpha_2 + \left(\frac{\xi^2 + \xi}{2}\right) \alpha_3 + \frac{2}{l} \left(\left(\xi - \frac{1}{2}\right) \omega_1 - 2\xi \omega_2 + \left(\xi + \frac{1}{2}\right) \omega_3 \right)$$

$$E(\xi) = \begin{bmatrix} K(\xi) \\ \gamma(\xi) \end{bmatrix} = \begin{bmatrix} 0 & \frac{2}{l} N_1' & 0 & \frac{2}{l} N_2' & 0 & \frac{2}{l} N_3' \\ \frac{2}{l} N_1' & -N_1 & \frac{2}{l} N_2' & -N_2 & \frac{2}{l} N_3' & -N_3 \end{bmatrix} \underline{u} = \underline{B} \underline{u}$$

$$= \begin{bmatrix} 0 & \frac{2\xi-1}{l} & 0 & -\frac{4}{l}\xi & 0 & \frac{2\xi+1}{l} \\ \frac{2\xi-1}{l} & -\frac{\xi}{2}(\xi-1) & \frac{-4\xi}{l} & \xi^2-1 & \frac{2\xi+1}{l} & -\frac{\xi}{2}(\xi+1) \end{bmatrix} \underline{u}$$

(iii) Constitutive,

$$K = \begin{bmatrix} EI & 0 \\ 0 & GA_s \end{bmatrix}$$

$$\sigma(\xi) = \underline{K} \underline{B} \underline{u}$$

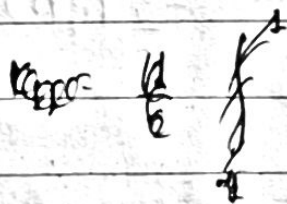
(iv) Element Stiffness Matrix,

$$K_e = \int_{-1}^1 \underline{B}^T \underline{K} \underline{B} d\xi = \frac{1}{2} \int_{-1}^1 \underline{B}^T \underline{K} \underline{B} d\xi$$

$$\underline{\underline{B^T K B}} = \begin{bmatrix} 0 & \frac{2\xi-1}{l} \\ \frac{2\xi-1}{l} & \frac{\xi}{2}(1-\xi) \\ 0 & -\frac{4\xi}{l} \\ -\frac{4\xi}{l} & \xi^2-1 \\ 0 & \frac{2\xi H}{l} \\ \frac{2\xi H}{l} & -\frac{\xi}{2}(\xi+1) \end{bmatrix} \begin{bmatrix} EI & 0 \\ 0 & GA_s \end{bmatrix} \begin{bmatrix} 0 & \frac{2\xi-1}{l} & 0 & -\frac{4\xi}{l} & 0 & \frac{2\xi H}{l} \\ \frac{2\xi-1}{l} & \frac{\xi}{2}(1-\xi) & -\frac{4\xi}{l} & \xi^2-1 & \frac{2\xi H}{l} & -\frac{\xi}{2}(\xi+1) \end{bmatrix}$$

$2 \times 2 \quad \quad \quad 2 \times 6$

6×2



$$\Rightarrow \underline{\underline{B^T K B}} = \begin{bmatrix} 0 & \frac{2\xi-1}{l} \\ \frac{2\xi-1}{l} & \frac{\xi}{2}(1-\xi) \\ 0 & -\frac{4\xi}{l} \\ -\frac{4\xi}{l} & \xi^2-1 \\ 0 & \frac{2\xi H}{l} \\ \frac{2\xi H}{l} & -\frac{\xi}{2}(\xi+1) \end{bmatrix} \begin{bmatrix} 0 & \frac{EI}{l}(2\xi-1) & 0 & -\frac{EI}{l}(4\xi) & 0 & \frac{EI}{l}(2\xi H) \\ \frac{GA_s}{l}(2\xi-1) & \frac{GA_s}{2}\xi(1-\xi) & -\frac{GA_s}{l}4\xi & GA_s(\xi^2-1) & \frac{GA_s}{l}(2\xi H) & -\frac{GA_s}{2}\xi(\xi+1) \end{bmatrix}$$

$$K_{11} = \frac{1}{2} \int_{-1}^1 \left(\frac{2\xi-1}{1} \right) \frac{GA_s}{1} (2\xi-1) d\xi = \frac{7}{3l} GA_s$$

$$K_{12} = \frac{1}{2} \int_{-1}^1 \left(\frac{2\xi-1}{1} \right) \frac{GA_s}{2} \xi(1-\xi) d\xi = \frac{GA_s}{2}$$

$$K_{13} = \frac{1}{2} \int_{-1}^1 \left(\frac{2\xi-1}{1} \right) \left(-\frac{GA_s}{2} \right) (4\xi) d\xi = -\frac{8}{3l} GA_s$$

$$K_{14} = \frac{1}{2} \int_{-1}^1 \left(\frac{2\xi-1}{1} \right) GA_s (\xi^2-1) d\xi = \frac{2GA_s}{3}$$

$$K_{15} = \frac{1}{2} \int_{-1}^1 \left(\frac{2\xi-1}{1} \right) \frac{GA_s}{1} (2\xi+1) d\xi = \frac{GA_s}{3l}$$

$$K_{16} = \frac{1}{2} \int_{-1}^1 \left(\frac{2\xi-1}{1} \right) \left(-\frac{GA_s}{2} \right) \xi(\xi+1) d\xi = -\frac{GA_s}{6}$$

$$K_{22} = \frac{1}{2} \int_{-1}^1 \left[\left(\frac{2\xi-1}{1} \right) \left(\frac{EI}{1} (2\xi-1) \right) + \frac{\xi(1-\xi)}{2} \frac{GA_s}{2} \xi(1-\xi) \right] d\xi = \frac{2EI}{3l} + \frac{2GA_sl}{15}$$

$$K_{23} = \frac{1}{2} \int_{-1}^1 \frac{\xi(1-\xi)}{2} \left(-\frac{GA_s}{1} \right) 4\xi d\xi = -\frac{2GA_sl}{3}$$

$$K_{24} = \frac{1}{2} \int_{-1}^1 \left[\left(\frac{2\xi-1}{1} \right) \left(-\frac{EI}{1} 4\xi \right) + \frac{\xi(1-\xi)}{2} GA_s (\xi^2-1) \right] d\xi = \frac{-8EI}{3l} + \frac{GA_sl}{15}$$

Similarly,

$$K_{25} = \frac{GA_3}{6}$$

$$K_{26} = \frac{EI}{3L} - \frac{GA_3 L}{30}$$

$$K_{33} = \frac{16GA_3}{3L}$$

$$K_{34} = 0$$

$$K_{35} = -\frac{8GA_3}{3L}$$

$$K_{36} = \frac{2GA_3}{3}$$

$$K_{44} = \frac{16EI}{3L} + \frac{8GA_3 L}{15}$$

$$K_{45} = -\frac{2GA_3}{3}$$

$$K_{46} = \frac{-8EI}{3L} + \frac{GA_3 L}{15}$$

$$K_{55} = \frac{7GA_3}{3L}$$

$$K_{36} = -\frac{GA_s}{2}$$

$$K_{66} = \frac{7EI}{3L} + \frac{2GA_s L}{15}$$

$$\therefore K_e =$$

	$\frac{7GA_s}{3L}$	$\frac{GA_s}{2}$	$-\frac{8GA_s}{3L}$	$\frac{2GA_s}{3}$	$\frac{GA_s}{3L}$	$-\frac{GA_s}{6}$
		$\frac{7EI}{3L} + \frac{2GA_s L}{15}$	$-\frac{2GA_s}{3}$	$-\frac{8EI}{3L} + \frac{GA_s L}{15}$	$\frac{GA_s}{6}$	$\frac{EI}{3L} - \frac{GA_s L}{30}$
			$\frac{16GA_s}{3L}$	0	$-\frac{8GA_s}{3L}$	$\frac{2GA_s}{3}$
				$\frac{16EI}{3L} + \frac{8GA_s L}{15}$	$-\frac{2GA_s}{3}$	$-\frac{8EI}{3L} + \frac{GA_s L}{15}$
					$\frac{7GA_s}{3L}$	$-\frac{GA_s}{2}$
						$\frac{7EI}{3L} + \frac{2GA_s L}{15}$

from analytical solution,

$$K \approx \frac{EI}{L^3(1+\phi)} \begin{bmatrix} 12 & 6L & -12 & 6L \\ (4+\phi)L^2 & -6L & (2-\phi)L^2 & \\ & 12 & -6L & \\ & & (4+\phi)L^2 & \end{bmatrix}, \quad \phi = \frac{12EI}{GA_s L^2}$$

The analytical matrix is exact, while Problem #1 and #2 are approximations, and they approximate the same thing with different assumed displacement fields.

In Problem #1 and #2, we guess independent polynomials, linear w and linear α in #1, quadratic w and quadratic α in #2. Then we calculate $K_e = \int B^T K B dx$. Those guessed fields don't satisfy

the Timoshenko ODEs, so the matrices are only approximate.

An assumed FE matrix equals the exact one only if the exact solution lives inside/lies in the interpolation space, and neither equal-order polynomial space contains the exact cubic w and quadratic α coupled solution.

$$K_{\#1} \geq K_{\#2} \geq K_{\text{analytical}}$$

Problem # 3

Analytical tip deflection: Timoshenko=0.030860 in

No of Elements	Length of element (in)	Tip deflection (in)	% of Timoshenko Deflection	Moment at fixed end (kip-in)
2	12	0.013523	43.8%	240
4	6	0.023370	75.7%	240
8	3	0.028570	92.6%	240

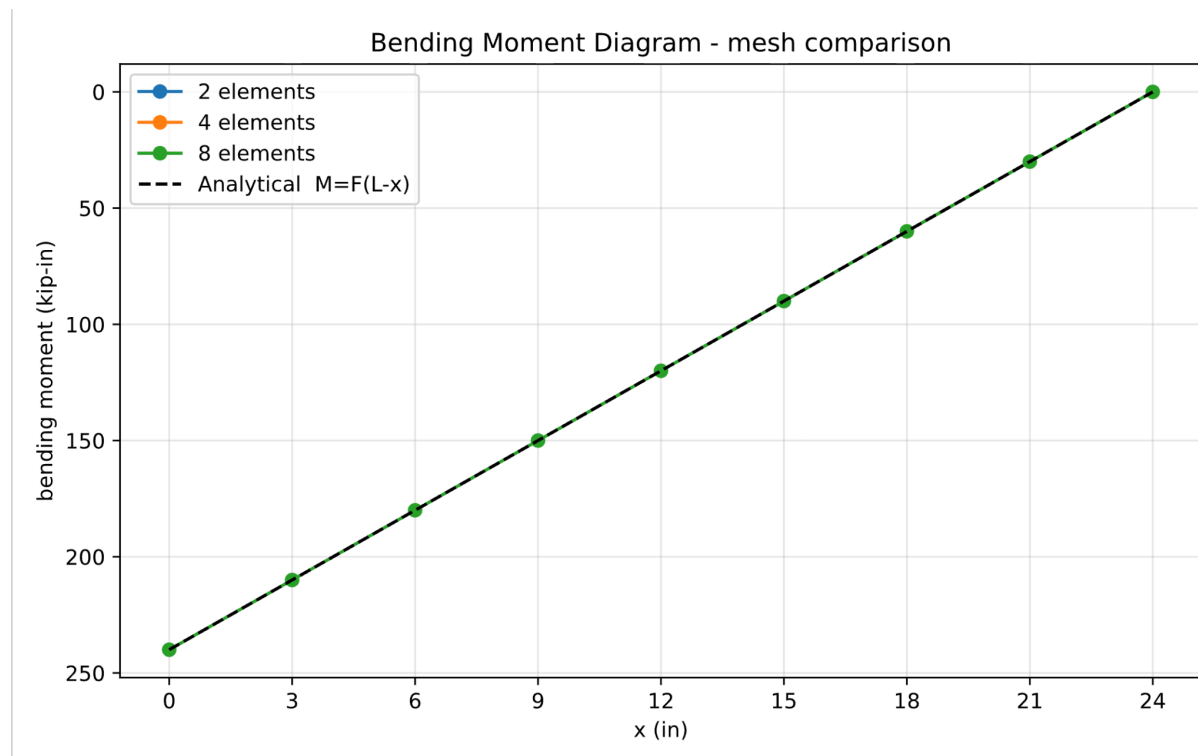


Fig. BMD for Analytical (Timoshenko) and Linear Displacement Field

BMD overlaps (all meshes + analytical coincide)

The applied load is a single tip force, so the true shear is constant ($V=F$) and the true moment is exactly linear, $M=F(L-x)$. The FE nodal moments come from $\{f_e\}=[K_e]\{d_e\}$, which enforces exact equilibrium at every node. So even though the deflection is affected by shear locking, each node reports the equilibrium moment, and those values are exactly $F(L-x)$ for all three meshes.

When we standardize Timoshenko beam elements (which account for both bending and shear deformation), the stiffness matrix couples bending and shear terms. For a thin beam under a tip load, the true behavior is almost purely flexural and shear deformation should be negligible.

But the element's shape functions can't independently represent bending curvature without also generating false shear strains. The element effectively becomes artificially stiff (shear locking) and under-predicting deflection.

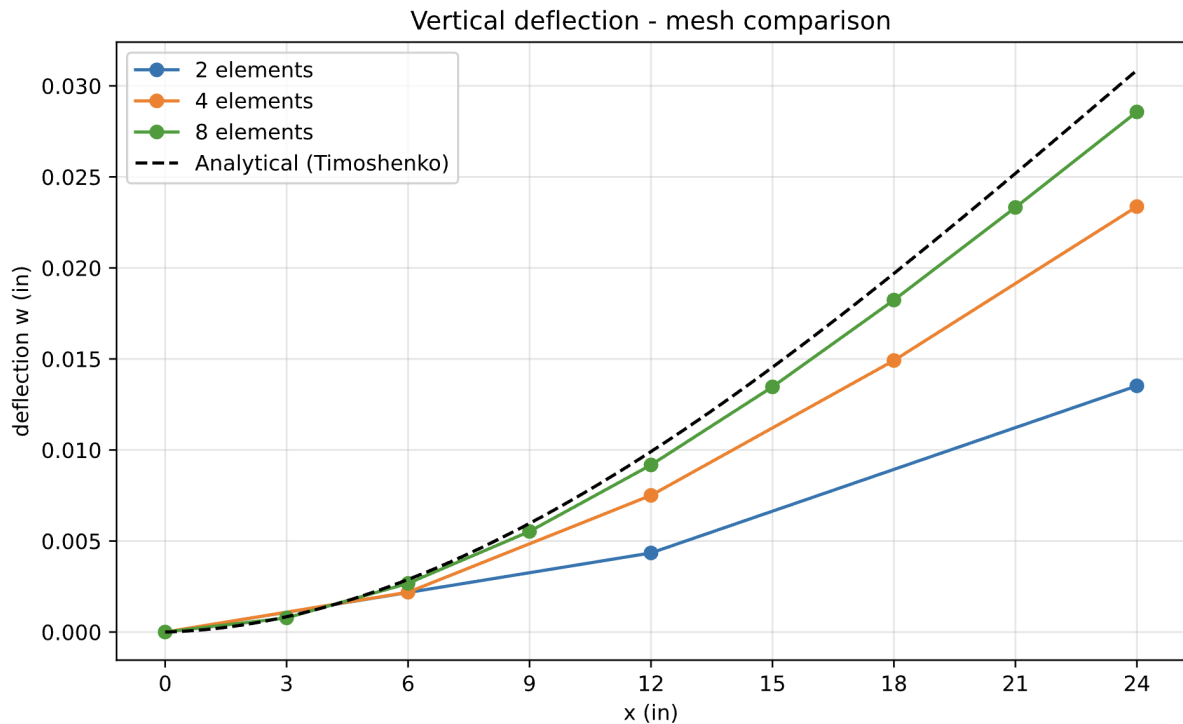


Fig. Vertical Deflection for Analytical (Timoshenko) and Linear Displacement Field

Deflection is strongly mesh-dependent,

- 1) 2 elements reach only 43.8% of the true tip value, 4 elements 75.7%, 8 elements 92.6%.
- 2) Curves converge upward (from below) as the mesh refines.

Observation:

The displacement-based linear element (Problem #1) is too stiff and underpredicts deflection, converging to the analytical value only as no of elements is increased.

It is because of shear locking: with linear w and linear α , the shear strain $\gamma = w' - \alpha$ can't vanish in pure bending, injecting false shear energy that scales as $\phi_e = (12 E I) / (G A_s L_e^2)$. Refining the mesh shrinks L_e and reduces this affect, so deflection improves monotonically toward 0.030860 in.

Python Script

Link: <https://github.com/nirajkyadav/Homeworks/blob/main/HW3/Q3.py>

Bonus Problem

In Problem #3, also use the 3-node Timoshenko element derived in Problem #2. Compare all the results and state your observations with reasons.

Analytical tip deflection: Timoshenko = 0.030860 in

Element	No of Elements	Length of element (in)	Tip deflection (in)	% of Timoshenko Deflection	Moment at fixed end (kip-in)
Linear (2-node)	2	12	0.013523	43.8%	240
Linear (2-node)	4	6	0.023370	75.7%	240
Linear (2-node)	8	3	0.028570	92.6%	240
Quadratic (3-node)	1	24	0.027135	87.9%	240

Observations:

One 3-node element (87.9%) beats the 4-element linear mesh (75.7%) and nearly matches the 8-element linear mesh (92.6%) by using only 6 DOF versus the 8-element mesh's 18 DOFs.

This is because the quadratic element carries a parabolic displacement fields (w and α) and , so its shear strain $\gamma = w' - \alpha$ (linear w' minus quadratic α) is far closer to the true cantilever state than the linear element's constant- w' and linear- α field.

Both element types underpredict deflection. The quadratic element still reaches only 87.9% with one element because the exact cantilever deflection is cubic in x , while the element field is only quadratic, and a residual shear-locking mismatch remains. Refining the quadratic mesh would significantly increase its closeness to true value.

The deflection plot shows the shape difference, not just the magnitude. The quadratic curve (dashed red) is a smooth parabola passing through its 3 nodes, tracking the analytical curvature closely through mid-span, whereas the linear meshes are piecewise-straight chords that visibly cut under the true curve.

The applied load is a single tip force, so the true shear is constant ($V=F$) and the true moment is exactly linear, $M=F(L-x)$. The FE nodal moments come from $\{f_e\} = [K_e]\{d_e\}$, which enforces exact equilibrium at every node. So even though the deflection is affected by shear locking, each node reports the equilibrium moment, and those values are exactly $F(L-x)$ for all three meshes.

Python Script

Link: https://github.com/nirajkyadav/Homeworks/blob/main/HW3/Bonus_Question.py

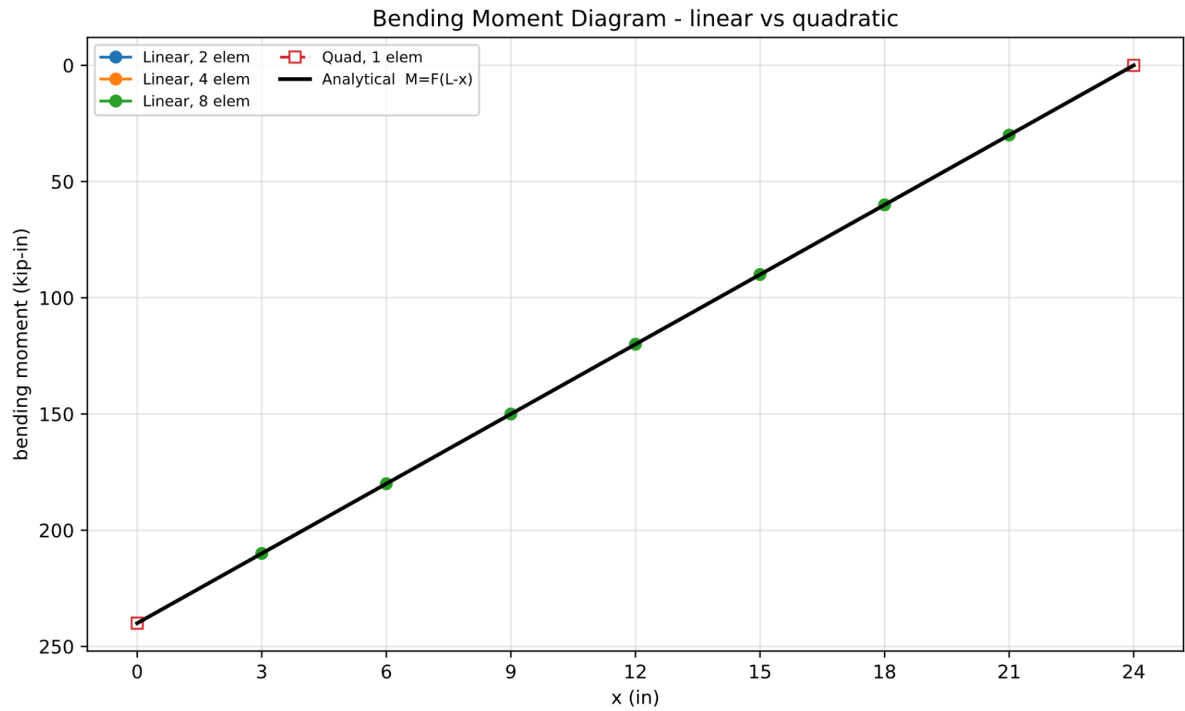


Fig. BMD for Analytical (Timoshenko), Linear & Quadratic Displacement Field

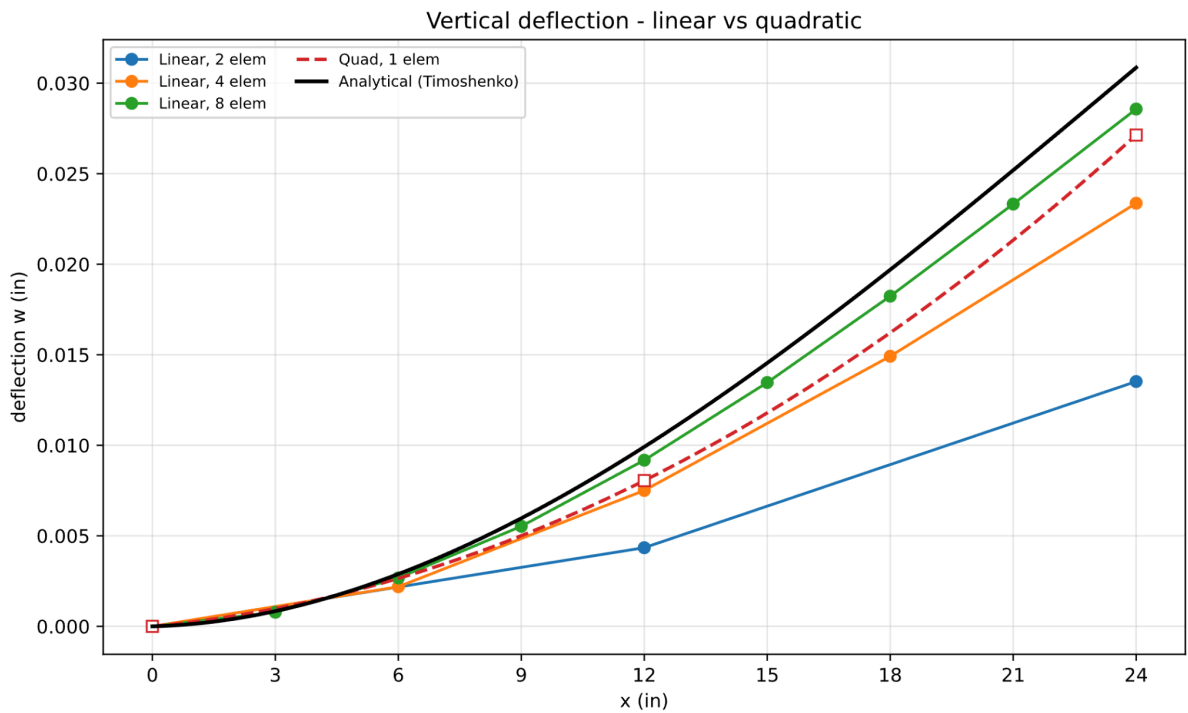


Fig. Vertical Deflection for Analytical (Timoshenko), Linear & Quadratic Displacement Field

③ Calculation of tip deflection using analytical solution

we have, $M(x) = EI \frac{d^2 \alpha(x)}{dx^2}$, $\frac{dy}{dx} = \alpha(x) = \frac{EI}{GA_s} \frac{d^2 \alpha}{dx^2}$

$$\alpha(x) = C_0 + C_1 x + C_2 x^2 + \frac{\omega_0}{6EI} x^3$$

$$y(x) = \left(C_0 - \frac{2EI}{GA_s} C_2 \right) x + \left(C_1 - \frac{\omega_0}{GA_s} \right) \frac{x^2}{2} + C_2 \frac{x^3}{3} + \frac{\omega_0}{6EI} \left(\frac{x^4}{4} \right) + C_3$$

$$\Rightarrow M(x) = EI \left(C_1 + 2x C_2 + \frac{\omega_0 x^2}{2EI} \right)$$

we have,

At $x=0$,

$$\Rightarrow V(x) = EI \left(2C_2 + \frac{\omega_0 x}{EI} \right)$$

$$y=0 \Rightarrow C_3=0$$

$$\alpha=0 \Rightarrow C_0=0$$

At $x=L$,

$$V(L) = F_{tip} = EI \left(2C_2 + \frac{\omega_0 L}{EI} \right)$$

$$\Rightarrow \frac{1}{2} \left(\frac{F_{tip}}{EI} - \frac{\omega_0 L}{EI} \right) = C_2 = \frac{1}{2EI} (F_{tip} - \omega_0 L)$$

$$M(L) = 0 = EI \left(C_1 + 2C_2 L + \frac{\omega_0 L^2}{2EI} \right)$$

$$\Rightarrow C_1 = -2L \cdot \frac{1}{2} \left(\frac{F_{tip}}{EI} - \frac{\omega_0 L}{EI} \right) - \frac{\omega_0 L^2}{2EI}$$

$$\Rightarrow C_1 = -\frac{F_{tip} L}{EI} + \frac{\omega_0 L^2}{EI} - \frac{\omega_0 L^2}{2EI}$$

$$= -\frac{F_{tip} L}{EI} + \frac{\omega_0 L^2}{2EI}$$

$$C_1 = \frac{1}{2EI} (\omega_0 L^2 - 2F_{tip} L)$$

$$\Rightarrow y(x) = \frac{-2EI}{GA_s} \cdot \frac{1}{2EI} (F_{tip} - w_0 L) x$$

$$+ \frac{1}{2EI} (w_0 L^2 - 2F_{tip} L) \frac{x^2}{2} - \frac{w_0}{GA_s} \frac{x^2}{2}$$

$$+ \frac{1}{2EI} (F_{tip} - w_0 L) \frac{x^3}{3}$$

$$+ \frac{w_0}{6EI} \frac{x^4}{4}$$

putting $w_0 = 0$,

$$y(x) = -\frac{F_{tip}}{GA_s} x - \frac{F_{tip} L}{2EI} x^2 + \frac{F_{tip}}{6EI} x^3$$

$$= \frac{F_{tip}}{2EI} \left(-Lx^2 + \frac{x^3}{3} \right) - \frac{F_{tip}}{GA_s} x$$

$$= -\frac{1}{6EI} \left[\frac{F_{tip} x^2 (3L - x)}{1} + \frac{F_{tip} x}{GA_s} \right]$$

Here,

$$\frac{dy}{dx} = -\frac{1}{6EI} \left(-2x(3L - x) + \frac{EI}{GA_s} \frac{d^2 \theta}{dx^2} \right)$$

$$\therefore \boxed{y(x) = \frac{F_{tip} x^2 (3L - x)}{6EI} + \frac{F_{tip} x}{GA_s}}$$